

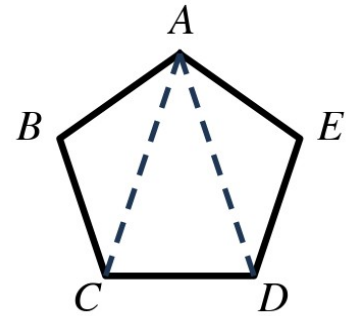
**Speak Like A Mathematician:**

polygon:

side · vertex · diagonal:

equilateral · equiangular · regular:

convex · concave:



**Polygon Angle Theorems**

**Polygon Interior Angles Theorem:** the sum of the interior angle measures of a convex n-gon is  $( \underline{\hspace{1cm}} - 2 ) \cdot \underline{\hspace{1cm}}^\circ$ .

**Polygon Exterior Angles Theorem:** the sum of the exterior angle measures of a convex polygon, one at each vertex, is  $\underline{\hspace{1cm}}^\circ$ .

### Example 1 - Name That Polygon

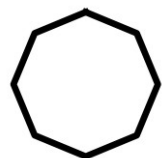
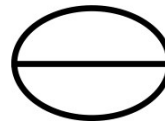
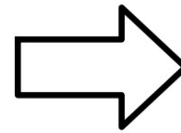
Fill in the name of each polygon by its number of sides.

| # of sides | Name | # of sides | Name |
|------------|------|------------|------|
| 3          |      | 8          |      |
| 4          |      | 9          |      |
| 5          |      | 10         |      |
| 6          |      | 12         |      |
| 7          |      | n          |      |

### Example 2 - Polygon or Not?

Tell whether each figure is a polygon. If it is, name it by its number of sides.

- a) Figures 1 and 2 at the right.
- b) Figures 3 and 4 at the right.
- c) Figures 5 and 6 at the right.



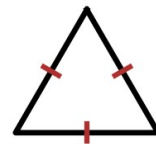
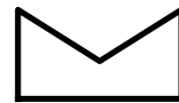
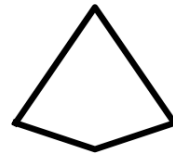
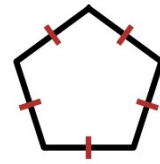
### Example 3 - Classify Each Polygon

Tell whether each polygon is regular or irregular, and concave or convex.

- a) Figures 1 and 2 at the right.
- b) Figures 3 and 4 at the right.
- c) Figures 5 and 6 at the right.

### Example 4 - Interior Angle Sums

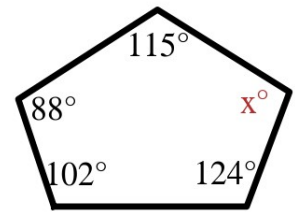
- a) Find the sum of the interior angle measures of a convex nonagon.
- b) Find the measure of each interior angle of a regular 20-gon.
- c) Find the sum of the interior angle measures of a convex 14-gon.



### Example 5 - Find the Value of x

The pentagon at the right has interior angles of  $102^\circ$ ,  $124^\circ$ ,  $x^\circ$ ,  $115^\circ$ , and  $88^\circ$ .

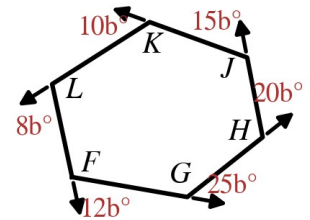
- What must the five angles add to? Write the equation.
- Solve for x.



### Example 6 - Exterior Angles: Find b

In hexagon FGHJKL at the right, the exterior angles measure  $12b^\circ$ ,  $25b^\circ$ ,  $20b^\circ$ ,  $15b^\circ$ ,  $10b^\circ$ , and  $8b^\circ$ .

- Write the equation and solve for b.
- Find the largest exterior angle.



### Example 7 - Exterior Angles: Find r

In quadrilateral JKLM at the right, the exterior angles measure  $3r^\circ$ ,  $8r^\circ$ ,  $5r^\circ$ , and  $4r^\circ$ .

- Find r.
- Find each exterior angle. Check that they total  $360^\circ$ .

